INSTRUCTOR'S DISCUSSION GUIDE

Topic 5: Network Cable Types and Media

Compare network cable types, categories, wiring standards, termination, and wire mapping.

Curriculum: Networking 1 **Excluded:** Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide) **Total Discussion Units:** 10 Slides

Slide 3 of 13 **Five network media choices** CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A transmission medium carries signals between network interfaces. Twisted-pair copper, fiber, coaxial cable, and radio differ in construction, reach, interference behavior, and installation needs.

- Unshielded Twisted Pair (UTP): Four unshielded twisted copper pairs, no overall shield. Twisting reduces electromagnetic interference (EMI).
- Shielded Twisted Pair (STP): Shielded twisted-pair cable with foil or braid shielding; requires proper grounding and bonding.
- Fiber: Carries light signals through glass or plastic cores; immune to EMI and spans long distances.
- Coaxial: Copper core with dielectric insulator and braided metallic shield; used for broadband and legacy bus.
- Wireless: Radio frequency transmission providing mobility across shared, unguided electromagnetic airspace.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

Compare network cable types, categories, wiring standards, termination, and wire mapping.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- UTP: Four unshielded twisted copper pairs, no overall shield.
- Twisting reduces electromagnetic interference (EMI).
- STP: Shielded twisted-pair cable with foil or braid shielding; requires proper grounding and bonding.
- Fiber: Carries light signals through glass or plastic cores; immune to EMI and spans long distances.
- Coaxial: Copper core with dielectric insulator and braided metallic shield; used for broadband and legacy bus.
- Wireless: Radio frequency transmission providing mobility across shared, unguided electromagnetic airspace.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of five network media choices.

Key Takeaway: A transmission medium carries signals between network interfaces. Twisted-pair copper, fiber, coaxial cable, and radio differ in construction, reach, interference behavior, and installation needs.

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Three Ethernet cable categories

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A cable category describes specified performance characteristics for compatible Ethernet installations. Supported rate and reach also depend on the standard, connectors, channel length, and installation quality.

- Cat 5e: Supports 1 Gb/s (1000BASE-T) over a standard 100 m channel.
- Cat 6: Supports 1 Gb/s at 100 m; supports 10 Gb/s (10GBASE-T) up to 55 m.
- Cat 6A: Supports 10 Gb/s (10GBASE-T) over a full 100 m channel.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Cat 5e: Supports 1 Gb/s (1000BASE-T) over a standard 100 m channel.
- Cat 6: Supports 1 Gb/s at 100 m; supports 10 Gb/s (10GBASE-T) up to 55 m.
- Cat 6A: Supports 10 Gb/s (10GBASE-T) over a full 100 m channel.
- A standard channel comprises up to 90 m of permanent horizontal link plus 10 m of patch cords.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of three ethernet cable categories.

Key Takeaway: A cable category describes specified performance characteristics for compatible Ethernet installations. Supported rate and reach also depend on the standard, connectors, channel length, and installation quality.

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RJ45 Plugs and Keystone Jacks

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Twisted-pair Ethernet uses Eight Positions, Eight Contacts (8P8C) modular plugs for patch cords and 110 Insulation-Displacement Contact (IDC) keystone jacks for permanent wall runs. Both components must match the cable category and wiring standard.

- Registered Jack 45 (RJ45) (Eight Positions, Eight Contacts (8P8C)) plug: Modular male plug with eight gold-plated contacts, used to terminate flexible stranded patch cables.
- Keystone Jack: Modular female jack with 110 Insulation-Displacement Contact (IDC) contacts, mounted in wall faceplates and patch panels for permanent solid-core horizontal cabling.
- Category matching: Plugs and jacks must match the cable category and conductor gauge to prevent signal degradation.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- RJ45 (8P8C) plug: Modular male plug with eight gold-plated contacts, used to terminate flexible stranded patch cables.
- Keystone Jack: Modular female jack with 110 IDC contacts, mounted in wall faceplates and patch panels for permanent solid-core horizontal cabling.
- Category matching: Plugs and jacks must match the cable category and conductor gauge to prevent signal degradation.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of rj45 plugs and keystone jacks.

Key Takeaway: Twisted-pair Ethernet uses Eight Positions, Eight Contacts (8P8C) modular plugs for patch cords and 110 Insulation-Displacement Contact (IDC) keystone jacks for permanent wall runs. Both components must match the cable category and wiring standard.

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T568A: eight conductor positions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

T568A specifies which conductor occupies each of the eight contact positions. Read the color order from pin 1 to pin 8 using the stated connector orientation.

- Pin 1: White/Green.
- Pin 2: Green.
- Pin 3: White/Orange.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Orange.
- Pin 7: White/Brown.
- Pin 8: Brown.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Pin 1: White/Green.
- Pin 2: Green.
- Pin 3: White/Orange.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Orange.
- Pin 7: White/Brown.
- Pin 8: Brown.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of t568a: eight conductor positions.

Key Takeaway: T568A specifies which conductor occupies each of the eight contact positions. Read the color order from pin 1 to pin 8 using the stated connector orientation.

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T568B: eight conductor positions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

T568B is another standardized eight-contact conductor arrangement. It exchanges the orange and green pair positions relative to T568A while preserving the blue and brown positions.

- Pin 1: White/Orange.
- Pin 2: Orange.
- Pin 3: White/Green.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Green.
- Pin 7: White/Brown.
- Pin 8: Brown.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Pin 1: White/Orange.
- Pin 2: Orange.
- Pin 3: White/Green.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Green.
- Pin 7: White/Brown.
- Pin 8: Brown.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of t568b: eight conductor positions.

Key Takeaway: T568B is another standardized eight-contact conductor arrangement. It exchanges the orange and green pair positions relative to T568A while preserving the blue and brown positions.

Slide 8 of 13

Straight-Through vs Crossover Cables

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A straight-through cable uses the same termination standard at both ends. A traditional crossover changes selected pair positions, although many modern interfaces can adjust automatically.

- A conventional A–B crossover swaps pins 1↔3 and 2↔6 for traditional 10/100BASE-T transmit/receive pairs.
- Many modern interfaces support auto-MDI/Medium Dependent Interface Crossover (MDIX), so a crossover cable is often unnecessary. Gigabit Ethernet uses all four pairs.
- An equal but nonstandard color order may pass a simple continuity check while creating split pairs and poor performance.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

Compare network cable types, categories, wiring standards, termination, and wire mapping.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A conventional A–B crossover swaps pins 1↔3 and 2↔6 for traditional 10/100BASE-T transmit/receive pairs.
- Many modern interfaces support auto-MDI/MDIX, so a crossover cable is often unnecessary.
- Gigabit Ethernet uses all four pairs.
- An equal but nonstandard color order may pass a simple continuity check while creating split pairs and poor performance.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of straight-through vs crossover cables.

Key Takeaway: A straight-through cable uses the same termination standard at both ends. A traditional crossover changes selected pair positions, although many modern interfaces can adjust automatically.

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Four cabling tool and tester functions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Cabling tools prepare, terminate, or test different parts of an installation. Stripping, crimping, punch-down termination, and continuity testing each require the appropriate tool.

- Cable stripper: removes jacket without nicking conductor insulation.
- Crimping tool: terminates compatible modular plugs.
- Punch-down tool: seats conductors in compatible Insulation-Displacement Contact (IDC) jacks or patch panels.
- Basic Local Area Network (LAN) tester: checks conductor continuity and wire mapping.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

Compare network cable types, categories, wiring standards, termination, and wire mapping.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Cable stripper: removes jacket without nicking conductor insulation.
- Crimping tool: terminates compatible modular plugs.
- Punch-down tool: seats conductors in compatible IDC jacks or patch panels.
- Basic LAN tester: checks conductor continuity and wire mapping.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of four cabling tool and tester functions.

Key Takeaway: Cabling tools prepare, terminate, or test different parts of an installation. Stripping, crimping, punch-down termination, and continuity testing each require the appropriate tool.

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RJ45 Cable Crimping Tools

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Terminating an Eight Positions, Eight Contacts (8P8C) modular plug requires preparing dedicated cutting, stripping, crimping, and testing tools alongside quality Cat 5e/Cat 6 media.

- Registered Jack 45 (RJ45) Modular Crimper: Heavy-duty ratcheting tool with precision Eight Positions, Eight Contacts (8P8C) dies to compress contacts into copper conductors and engage the jacket strain relief.
- Unshielded Twisted Pair (UTP) Cable Stripper: Compact rotary wire stripper calibrated to cleanly score and remove the outer PVC jacket without nicking conductor insulation.
- Modular Registered Jack 45 (RJ45) Plugs: Cat 5e or Cat 6 clear polycarbonate male Eight Positions, Eight Contacts (8P8C) plugs featuring eight gold-plated contact pins and a flexible locking tab.
- Unshielded Twisted Pair (UTP) Cable: Four-pair 24 AWG stranded copper Ethernet cable designed for flexible patch cords between workstations and wall jacks.
- Flush Cutters / Snips: Precision diagonal cutting pliers used to trim untwisted, aligned wires straight and square across before insertion into the plug.
- Local Area Network (LAN) Cable Tester: Dual-unit continuity wire-map tester to confirm all eight pin paths and detect open circuits, short circuits, or crossed pairs.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- RJ45 Modular Crimper: Heavy-duty ratcheting tool with precision 8P8C dies to compress contacts into copper conductors and engage the jacket strain relief.
- UTP Cable Stripper: Compact rotary wire stripper calibrated to cleanly score and remove the outer PVC jacket without nicking conductor insulation.
- Modular RJ45 Plugs: Cat 5e or Cat 6 clear polycarbonate male 8P8C plugs featuring eight gold-plated contact pins and a flexible locking tab.
- Unshielded Twisted Pair (UTP) Cable: Four-pair 24 AWG stranded copper Ethernet cable designed for flexible patch cords between workstations and wall jacks.
- Flush Cutters / Snips: Precision diagonal cutting pliers used to trim untwisted, aligned wires straight and square across before insertion into the plug.
- LAN Cable Tester: Dual-unit continuity wire-map tester to confirm all eight pin paths and detect open circuits, short circuits, or crossed pairs.
- Always prepare spare 8P8C plugs; modular plugs cannot be reused once crimped.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of rj45 cable crimping tools.

Key Takeaway: Terminating an Eight Positions, Eight Contacts (8P8C) modular plug requires preparing dedicated cutting, stripping, crimping, and testing tools alongside quality Cat 5e/Cat 6 media.

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Keystone Jack Punch-Down Tools

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Terminating a permanent wall jack requires specialized insulation displacement tools, punch-down supports, and rated solid-core cabling.

- 110 Impact Punch-Down Tool: Spring-loaded impact driver with a reversible 110/66 blade that seats conductors into Insulation-Displacement Contact (IDC) V-notches and cuts excess wire flush.
- Keystone Jack (110 Insulation-Displacement Contact (IDC)): Registered Jack 45 (RJ45) female modular jack with 110 insulation displacement contacts, universal T568A/T568B color coding, and snap-in faceplate clips.
- Unshielded Twisted Pair (UTP) Cable Stripper: Round jacket stripper used to remove 1 to 1.5 inches of outer jacket while preserving pair twists up to the Insulation-Displacement Contact (IDC) blocks.
- Solid-Core Unshielded Twisted Pair (UTP) Cable: Four-pair 23/24 AWG solid copper horizontal cable engineered for structured in-wall and ceiling cable runs.
- Punch-Down Palm Tool / Stand: Contoured palm puck or base holder that cradles the jack securely to prevent hand injury and ensure even downward punching pressure.
- Insulation-Displacement Contact (IDC) Dust Caps & Cable Tester: Snap-on protective dust covers to lock conductors firmly in place, plus a Local Area Network (LAN) tester with patch cord to verify end-to-end pin mapping.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 110 Impact Punch-Down Tool: Spring-loaded impact driver with a reversible 110/66 blade that seats conductors into IDC V-notches and cuts excess wire flush.
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- Solid-Core UTP Cable: Four-pair 23/24 AWG solid copper horizontal cable engineered for structured in-wall and ceiling cable runs.
- Punch-Down Palm Tool / Stand: Contoured palm puck or base holder that cradles the jack securely to prevent hand injury and ensure even downward punching pressure.
- IDC Dust Caps & Cable Tester: Snap-on protective dust covers to lock conductors firmly in place, plus a LAN tester with patch cord to verify end-to-end pin mapping.
- Always orient the punch-down tool with the CUT blade facing outward toward the excess wire tail.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of keystone jack punch-down tools.

Key Takeaway: Terminating a permanent wall jack requires specialized insulation displacement tools, punch-down supports, and rated solid-core cabling.

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Three common wire-map faults

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A wire map identifies which contact paths connect between cable ends. Opens, shorts, and miswires describe different continuity problems and require different corrections.

- Open: a conductor has no continuous end-to-end path.
- Short: conductors are accidentally joined.
- Miswire: a conductor arrives at an unexpected pin.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Compare network cable types, categories, wiring standards, termination, and wire mapping.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Open: a conductor has no continuous end-to-end path.
- Short: conductors are accidentally joined.
- Miswire: a conductor arrives at an unexpected pin.
- A wire-map pass does not certify pair integrity, signal quality, or cable bandwidth.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of three common wire-map faults.

Key Takeaway: A wire map identifies which contact paths connect between cable ends. Opens, shorts, and miswires describe different continuity problems and require different corrections.